Heterogeneous Returns to Antenatal Care: Evidence from Childhood Immunization in Senegal

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April 2026

Abstract

Antenatal care (ANC) serves both as a medical input and as a source of health information in the child health production function. We use seven waves of the Senegal Demographic and Health Surveys, matched to regional health-facility data, to estimate the effect of ANC adherence on childhood vaccination outcomes. We address endogeneity arising from unobserved heterogeneity in ANC utilization using a two-stage copula-based control-function approach. We find that ANC adherence has no effect on the composite measure of full immunization, but significantly increases uptake of two injectable vaccines, the three-dose diphtheria-pertussis-tetanus vaccine (DTP3) and measles, by 8.2 and 6.3 percentage points, respectively. These effects are concentrated among mothers with no schooling or only primary education. We also find no robust evidence that the marginal returns to ANC adherence differ by child sex, fertility intention, or ethnicity.

JEL classification: I12, J13, O15, C50

Keywords: antenatal care; child immunization; maternal education; heterogeneous effects; information frictions; Senegal.

Accepted manuscript of an article published in Economics & Human Biology (2026).

https://doi.org/10.1016/j.ehb.2026.101608

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Highlights:

- We analyze the relationship between antenatal care (ANC) adherence and child vaccination outcomes using nationally representative data from Senegal.
- ANC adherence has no effect on full immunization, but significantly increases uptake of DTP3 and measles.
- These positive effects are concentrated among less-educated mothers.
- Improving ANC adherence and counselling may complement immunization programs by strengthening information and trust, particularly among less educated mothers and for injectable vaccines.

1. Introduction

The economic returns to childhood vaccination extend far beyond health. Immunization not only reduces child mortality and morbidity (Shattock et al., 2024), but also protects households against catastrophic health expenditures and contributes to poverty reduction in low- and middle-income countries (Campbell et al., 2014; Ozawa et al., 2017; Jiao, Sato et al., 2025). By lowering exposure to large and uncertain medical costs, vaccination represents a high-return investment in human capital and financial protection. Yet despite these returns, childhood immunization remains incomplete for many children, and progress has stagnated across much of the developing world (Dubé et al., 2021; O'Brien and Lemango, 2023).

Since the infant health production models of Rosenzweig and Schultz (1982, 1983), building on Grossman's (1972) framework of health demand and health production, child health outcomes have been understood as the product of parents' early-life investments in human capital. Within this framework, antenatal care (ANC) is not only a medical input but also an informational one that may alter parental knowledge, beliefs, trust, and preventive behavior. A growing literature therefore treats ANC as an early-life parental investment with potential effects on postnatal child health outcomes (e.g., Alexander and Korenbrot, 1995; Currie and Grogger, 2002; Wehby et al., 2009; Habibov and Fan, 2011; Corman et al., 2019; Dubé et al., 2021; Cygan-Rehm and Karbownik, 2022), including childhood vaccination (Choi and Lee, 2006; Dixit et al., 2013; Krishnamoorthy and Rehman, 2022; Tiruneh et al., 2025).

Existing evidence documents a positive association between ANC visits and child immunization in low- and middle-income countries. Most studies rely on multivariate logit or probit models with socioeconomic and demographic controls (Shrivastwa et al., 2015; Mbengue et al., 2017; Noh et al., 2018; Sarker et al., 2019; Aalemi et al., 2020; Nour et al., 2020; Budu et al., 2021; Fenta et al., 2021; Barrow et al., 2023; Farrenkopf et al., 2023; Smalley et al., 2023). A few studies[2] use propensity score matching to address selection on observables (Dixit et al., 2013; Krishnamoorthy and Rehman, 2022; Tiruneh et al., 2025). Using Indian data, Dixit et al. (2013) and Krishnamoorthy and Rehman (2022) show that mothers with at least one ANC visit and, respectively, four or more visits are more likely to fully vaccinate their children than mothers with no ANC visits. Using Ethiopian data, Tiruneh et al. (2025) similarly report higher rates of full immunization among children whose mothers completed four or more ANC visits relative to those with fewer visits. However, to our knowledge, there is no evidence on the

[2] These studies rely on a single cross-sectional Demographic and Health Survey wave from one country and do not exploit variation across survey waves or local healthcare supply.

effect of ANC adherence on childhood immunization outcomes that accounts for selection on unobservables and distinguishes vaccine-specific effects. Unobserved factors such as maternal motivation, autonomy, and health-seeking preferences are likely to shape both ANC adherence and vaccination completion. Mothers who are more motivated or face lower access costs may be more likely both to adhere to ANC visits and to vaccinate their children, biasing estimates upward (Rosenzweig and Schultz, 1983; Choi and Lee, 2006). Conversely, pregnancy complications may increase ANC utilization while reducing subsequent compliance with postnatal care, biasing estimates downward (Conway and Deb, 2005).

We propose a conceptual framework in which ANC adherence, defined as completion of the WHO-recommended four visits, serves as an endogenous informational and trust-building input into childhood vaccination outcomes. Because vaccines differ in timing (early versus later in infancy), delivery modality (oral versus injectable), and the effort required for completion (single-dose versus multi-dose), and because parental preferences and beliefs shape child health investments (Ben-Porath and Welch 1976; Joyce et al. 2000; Jayachandran and Kuziemko 2011; Bharadwaj and Lakdawala 2013; Palloni 2017; Aurino 2017; Zerfu et al. 2009; Arriola and Grossman 2021), the returns to ANC may vary across vaccines and households. We test this framework using seven waves of nationally representative Demographic and Health Surveys from Senegal, merged with regional health-facility data. To address endogeneity arising from unobserved heterogeneity, we implement a copula-based control-function approach (2sCOPE) with generalized residuals, which is well suited to the binary nature of ANC adherence (Gourieroux et al. 1987; Petrin and Train 2010; Wooldridge 2015). Our baseline specification relies on a Gaussian copula (Trivedi and Zimmer 2007; Park and Gupta 2012; Yang et al. 2024), and we assess robustness to alternative copula forms.

We find that treating ANC adherence as exogenous yields broadly positive effects across vaccination outcomes. Once we account for unobserved heterogeneity, however, the pattern becomes more selective: ANC adherence has no effect on the composite measure of full immunization, but increases uptake of two injectable vaccines, the three-dose diphtheria-pertussis-tetanus vaccine (DTP3) and the single-dose measles vaccine. These effects are concentrated among mothers with no schooling or only primary education, consistent with an informational channel. We also find weaker and statistically insignificant estimates among urban girls for DTP3, and among unintended births and Poular/Peul households for both DTP3 and measles, although formal tests do not reject equality of effects across groups. Overall, our results suggest that policies aimed at improving ANC adherence can complement immunization programs through information and trust channels, with the largest gains for injectable vaccines.

The remainder of the paper is organized as follows. Section 2 describes the data and variables. Section 3 outlines the empirical strategy. Section 4 presents and discusses the results. Section 5 reports robustness checks, and Section 6 concludes.

2. Data and Variables

2.1. Data

To estimate the effect of ANC adherence on child vaccination, we use seven waves of the Senegal Demographic and Health Surveys (DHS) from 2012 to 2019[3]. The DHS is a publicly available, repeated cross-sectional survey covering nationally representative sample of women aged 15–49 and their children. We rely on the individual women's questionnaire, which contains detailed information on socio-demographic characteristics, family planning, contraception, and maternal and child health. In Senegal, the DHS is conducted by the National Agency of Statistics and Demography (ANSD), with technical and financial support from USAID[4], the World Bank, UNICEF, WHO, and UNFPA[5]. It serves as our primary data source for this study. We complement the DHS with regional measures of local health infrastructure, specifically the availability of hospitals, health centers, and health posts, drawn from ANSD regional health reports[6]. These indicators are merged to the DHS using administrative region identifiers.

The analysis focuses on children aged 12–23 months, the WHO-recommended age range for assessing full vaccination coverage. By this age, children are expected to have received all essential vaccines against major infectious diseases. To avoid within-mother correlation of observations, we retain only one child per mother[7], with the firstborn selected when multiple children fall within the eligible age range (e.g., twins). This yields an initial sample of 8,265 mother–child pairs. To preserve meaningful variation in ANC utilization while limiting the influence of outliers, we restrict the sample to mothers reporting at most eight ANC visits and

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[3] The 2014 DHS wave is excluded due to missing data on several key variables. We focus on the pre–COVID-19 period; earlier waves are not used because regional health infrastructure data required for matching are not available.

[4] United States Agency for International Development (USAID)

[5] United Nations Population Fund (UNFPA)

[6] This dataset is available from the author upon request.

[7] We control for the child's birth order and the total number of children in the household. We also assess the sensitivity of our results to this sampling restriction in Online Appendix D (Tables D2 and D3).

ANC initiation between the first and seventh month of pregnancy. These restrictions are motivated by both measurement and interpretation concerns at the extremes. Reports of ANC initiation in month zero are unlikely to reflect standard ANC practices (possibly reflecting measurement error or atypical timing, such as pre-pregnancy contacts or visits for pregnancy testing), whereas initiation in month eight or later is more plausibly associated with emergency- or complication-driven care than with routine ANC. Similarly, very high numbers of ANC visits are more likely to proxy for pregnancy complications. We refer to the resulting subsample as the full sample and examine the sensitivity of our main findings to these restrictions in the robustness analysis (see Online Appendix D). Fig. A1 in Online Appendix A plots the distributions of ANC initiation month and total ANC visits prior to imposing sample restrictions and indicates widespread compliance with official ANC guidelines. Detailed information on sample size before and after restrictions, by survey wave, is provided in Online Appendix A, Table A1.

To explore heterogeneity by maternal education, we construct three subsamples: (i) women who never attended school (no formal education), (ii) women with primary education, and (iii) women with secondary or higher education.

2.2. Variables

The DHS provides detailed information on childhood immunizations, including all required vaccines and doses for a child to be classified as fully vaccinated. A child is considered fully vaccinated if they have received: (i) one dose of the Bacillus Calmette–Guérin (BCG) vaccine, (ii) three doses of diphtheria–tetanus–pertussis (DTP) vaccine, (iii) three doses of polio vaccine (excluding the dose at birth), and (iv) one dose of measles vaccine. Official vaccination schedules and dose requirements are summarized in Online Appendix A, Table A2. For each child, we construct vaccine-specific binary indicators as well as a composite indicator equal to one if the child received all required doses and zero otherwise. This composite measure of full vaccination constitutes our primary outcome variable. To examine heterogeneity across vaccines, we also analyze vaccine-specific outcomes. As a placebo outcome, we additionally examine receipt of the birth dose of polio vaccine (Polio 0). Vaccination data are primarily obtained from vaccination cards presented by the mother at the time of the interview. In cases where the vaccination card is unavailable, maternal recall is used as a secondary source (Langsten & Hill, 1998). Temporal trends and regional disparities in vaccine-specific coverage and full immunization are reported in Online Appendix A (Tables A3–A7 and Figure A2).

While national coverage has increased over time, large and persistent regional disparities remain. In particular, the post-2017 increase in national coverage appears to reflect gains in some regions (e.g., Tambacounda and Kédougou) offsetting stagnation or declines in others, rather than broad-based progress across regions.

Table 1 reports descriptive statistics for antenatal care utilization and child vaccination outcomes. Approximately 60 percent of mothers initiate ANC by the third month of pregnancy, and 52 percent complete the four WHO-recommended ANC visits. Among early initiators, around 70 percent fully adhere to the recommended ANC schedule, compared with roughly one-quarter among late initiators. Vaccination rates are consistently higher among children born to ANC-adherent mothers across all vaccines, indicating a positive correlation between ANC adherence and child immunization outcomes.

Table 1. Descriptive statistics

Variable	Categories	Mean	N	Vaccination Rate				
				CV	*BCG*	*Polio3*	*DTP3*	*Measles*
ANC Adherence	yes	0.523	4,006	0.750	0.965	0.836	0.936	0.868
	no		3,654	0.696	0.937	0.806	0.866	0.792
Early ANC (by 3rd month)	yes	0.603	4,622	0.734	0.957	0.828	0.920	0.848
	no		3,038	0.709	0.943	0.813	0.876	0.807
Early ANC × Adherence	yes/yes	0.703	3,251	0.751	0.964	0.838	0.940	0.871
	yes/no		1,371	0.693	0.941	0.805	0.872	0.794
	no/yes	0.249	775	0.744	0.968	0.830	0.915	0.856
	no/no		2,283	0.697	0.934	0.807	0.862	0.791

Note: CV denotes full vaccination. ANC denotes antenatal care. ANC adherence equals one if the mother completed the four WHO-recommended ANC visits during pregnancy. Early ANC equals one if ANC was initiated by the WHO-recommended third month of pregnancy. BCG is a single-dose vaccine; Polio3 and DTP3 indicate receipt of three doses; Measles indicates receipt of one dose. Among mothers who initiated ANC early but did not adhere to the full schedule (29.7 percent), approximately 75 percent nevertheless completed at least three visits.

In the next section, we present the empirical strategy used to address unobserved heterogeneity and to estimate the effect of ANC adherence on child vaccination.

3. Empirical Strategy

3.1. Conceptual Framework

We conceptualize the link between maternal antenatal care (ANC) adherence and child vaccination completion as operating through three stages. In the first stage, the mother initiates ANC, marking her first contact with the health system during pregnancy. The timing of initiation is not random and is shaped by both structural and individual factors.

In the second stage, mothers who initiate ANC within the WHO-recommended first trimester (up to 12 weeks of gestation) have more time to comply with the recommended schedule of four ANC visits, given fixed pregnancy duration and recommended spacing of visits (Tunçalp et al., 2017; Jiwani et al., 2020; Asmare et al., 2025). The likelihood of adherence therefore depends on a calendar constraint common to all mothers, as well as on demand-side factors and supply-side factors. Earlier initiation mechanically expands the time available to complete four visits, thereby increasing the probability of adherence[8]. We therefore model ANC adherence as a function of initiation within the first trimester, demand-side factors, and supply-side factors. Omitting this feasibility component substantially reduce the predictive content of the adherence equation (Online Appendix B, Table B1) and may yield a noisier first-stage prediction of adherence.

In the third stage, the counselling and health education received during ANC visits may affect maternal beliefs, knowledge, trust in vaccines and in the health system, and perceptions of the benefits of immunization (Návar et al., 2007; Saitoh et al., 2013; Bangura et al., 2020; Dubé et al., 2021; Cygan-Rehm and Karbownik, 2022; Dasgupta et al., 2023; Unfried and Priebe, 2024). These informational inputs may, in turn, translate into postnatal child health investments, including vaccination completion. We therefore model child vaccination outcomes as a function of ANC adherence, demand-side factors, and supply-side factors[9]

These stages motivate the empirical strategy outlined in Section 3.2 and formalized in equations (1) and (2). Our objective is to estimate the effect of ANC adherence on full immunization and to assess heterogeneity across vaccines and maternal education. We also examine how household preferences, proxied by fertility intentions at conception and gender preference at birth, and beliefs, proxied by ethnicity, shape the returns to ANC adherence.

3.2. Econometric Model

We estimate the effect of antenatal care (ANC) adherence on the probability that a child completes the recommended vaccination schedule. For clarity, we present the model using full vaccination as the outcome; the same procedure applies to each of the four vaccine-specific outcomes.

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[8] Online Appendix B, Table B1 suggests that socioeconomic and demographic characteristics, as well as local healthcare availability, explain little of the variation in ANC adherence.

[9] Because ANC initiation and ANC adherence are closely related dimensions of ANC utilization (Tunçalp et al. 2017; Asmare et al. 2025), specifications that include both variables in the vaccination equation provide limited leverage for separately identifying their effects and yield unstable estimates (Online Appendix B, Table B2). We therefore assume that, conditional on observed demand- and supply-side factors as well as region and survey-wave fixed effects, any residual effect of initiating ANC by the third month operates through ANC adherence. Section 3.3 presents an estimator designed to strengthen the plausibility of this maintained restriction, which we further probe using birth-dose placebo outcomes (Online Appendix D, Table D5) and additional control variables (Online Appendix D, Table D4).

Let CV_i^* denote the latent propensity for child i to be fully vaccinated. Consistent with the conceptual framework described above, we specify:

$$CV_i^* = ANC_i \cdot \gamma + X_i \cdot \beta_1 + S_i \cdot \delta_1 + \lambda_r + \tau_t + \varepsilon_i, \qquad (1)$$

where ANC_i is a binary indicator equal to one if the mother of child i adhered to the WHO-recommended four ANC visits during pregnancy; X_i is a vector of maternal, child, partner, and household characteristics capturing demand-side factors; S_i captures supply-side factors related to local health care availability; λ_r denotes region fixed effects; τ_t denotes survey-wave fixed effects; and ε_i represents unobserved determinants of child vaccination completion. The parameter γ is the coefficient of interest.

The latent variable CV_i^* is not observed. Instead, we observe the binary outcome:

$$CV_i = \begin{cases} 0 \ si \ CV_i^* \leq 0 \\ 1 \ si \ CV_i^* > 0 \end{cases}$$

Similarly, let ANC_i^* denote the latent propensity of the mother to adhere to the recommended ANC schedule. We assume:

$$ANC_i^* = X_i \cdot \beta_2 + W_i \cdot \alpha + S_i \cdot \delta_2 + \lambda_r + \tau_t + v_i, \qquad (2)$$

where X_i, S_i, λ_r and τ_t are defined as above; W_i captures feasibility-related calendar constraints; and v_i represents unobserved determinants of ANC adherence. The variable W_i is measured by a binary indicator equal to one if ANC was initiated within the WHO-recommended first trimester of pregnancy. The observed ANC adherence indicator is defined as:

$$ANC_i = \begin{cases} 0 \ si \ ANC_i^* \leq 0 \\ 1 \ si \ ANC_i^* > 0 \end{cases}$$

The vector X_i includes a rich set of controls. Maternal characteristics include age, education, ethnicity, occupational status, and media exposure, defined as a binary indicator equal to one if the mother reports exposure to at least one media source (radio, television, or newspapers). Child-level controls include birth order. Partner characteristics are captured through occupational status. Household characteristics include urban versus rural residence, the DHS wealth index, relationship to the household head (wife, daughter or daughter-in-law, or other), number of children, and an indicator for polygamous households. The vector S_i include the number of hospitals, health centers, and health posts per 100,000 inhabitants at the regional level. Descriptive statistics for all control variables are reported in Online Appendix A.

The parameter γ in equation (1) captures the effect of antenatal care adherence on vaccination completion. Estimating equation (1) using a single-equation probit model may yield biased estimates if ANC_i is endogenous, as suggested by prior work (Choi and Lee, 2006; Habibov and Fan, 2011; Cygan-Rehm and Karbownik, 2022). Unobserved factors, such as maternal motivation, health preferences, or attitudes toward preventive care, may jointly affect ANC adherence and child vaccination completion and may not be fully captured by observed controls, thereby inducing correlation between ANC_i and the structural error term ε_i. If these same latent factors are also correlated with W_i, they may further weaken the plausibility of the maintained conditional restriction. We next describe the estimator we use to address this endogeneity concern and bolster the plausibility of that assumption.

3.3. A Copula-Based Control Function Approach: the 2sCOPE Estimator

We implement the 2sCOPE estimator proposed by Yang et al. (2024), which is conceptually related to the control-function approach (Terza et al., 2008; Petrin and Train, 2010; Wooldridge, 2015) and proceeds in two stages. We refer the reader to Yang et al. (2024) for the full technical development and focus here on how the estimator is adapted to a binary outcome with a binary endogenous regressor.

In the first stage, we estimate equation (2) using a probit model and compute the generated regressor, namely the generalized residual following Gourieroux et al. (1987):

$$\hat{\xi}_i = \frac{\phi(Z_i\,\hat{\theta})}{\Phi(Z_i\,\hat{\theta})\,(1-\Phi(Z_i\,\hat{\theta}))} \times [\,ANC_i - \Phi(Z_i\,\hat{\theta})]$$

where $Z_i = (X_i, W_i, S_i, \lambda_r, \tau_t)$, $\hat{\theta}$ denotes the vector of first-stage estimates, and $\phi(\cdot)$ and $\Phi(\cdot)$ denote the standard normal density and cumulative distribution functions.

More explicitly, the generalized residual can be written as a function of the observed value of ANC_i as:

$$\hat{\xi}_i = \begin{cases} \dfrac{\phi(Z_i\,\hat{\theta})}{\Phi(Z_i\,\hat{\theta})} & if\ ANC_i = 1 \\ -\dfrac{\phi(Z_i\,\hat{\theta})}{1-\Phi(Z_i\,\hat{\theta})} & if\ ANC_i = 0 \end{cases}$$

We next assume that the structural error term in the vaccination equation can be expressed as:

$$\varepsilon_i = \hat{\xi}_i \cdot \omega_1 + u_i,$$

where the control-function term $\hat{\xi}_i \cdot \omega_1$ captures the endogenous component of ANC adherence driven by latent factors such as maternal motivation, autonomy, and other unobserved health-seeking preferences. Including this term in the second-stage vaccination equation addresses the potential endogeneity of ANC_i (Terza et al., 2008; Wooldridge, 2015). The informativeness of the generated regressor $\hat{\xi}_i$ hinges on correct first-stage specification.

In the second stage, unlike the standard control-function approach, we allow for remaining dependence between the unobservables v_i and u_i, after conditioning on the generalized residual $\widehat{\xi}_i$. This accounts for potential approximation error in the generated regressor and for residual dependence beyond the linear control-function term. We specify their joint distribution using a Gaussian copula:

$$F_{(v_i,\ u_i)} = \mathbb{C}^{Gaussian}\,[F_{v_i}(v_i), F_{u_i}(u_i);\ \rho]\ ,$$

where $F_{v_i}(\cdot)$ and $F_{u_i}(\cdot)$ denote the marginal distributions of v_i and u_i, and ρ is the copula dependence parameter capturing residual correlation.

The resulting joint system estimated in the second stage can be written as:

$$\begin{cases} ANC_i^* = X_i \cdot \beta_2 + W_i \cdot \alpha + S_i \cdot \delta_2 + \lambda_r + \tau_t + v_i \\ CV_i^* = ANC_i \cdot \gamma + X_i \cdot \beta_1 + S_i \cdot \delta_1 + \lambda_r + \tau_t + \widehat{\xi}_i \cdot \omega_1 + u_i \end{cases} \quad (3)$$

We estimate system (3) by maximum likelihood using the GJRM package in R (Marra and Radice, 2019). The identifying assumption is that, conditional on observed demand- and supply-side factors, region and survey-wave fixed effects, the control-function term, and the copula dependence parameter, W_i has no residual direct effect on CV_i other than through ANC_i. To the extent that unobserved latent factors jointly shaping ANC initiation, ANC adherence, and child vaccination are absorbed by the copula and control-function terms, the 2sCOPE estimator consistently identifies γ (Yang et al., 2024; Haschka, 2025; Qian and Xie, 2025). Under this maintained assumption, γ can be interpreted as the causal effect of completing four ANC visits on vaccination completion, identified from adherence variation induced by calendar feasibility constraints linked to WHO-recommended first-trimester initiation. We assess the plausibility of this conditional restriction using birth-dose vaccines (BCG and Polio 0) as placebo outcomes in the subsample of facility births, where ANC adherence should not affect vaccination[10]. We find no statistically significant effects for either outcome in these placebo tests (Online Appendix D, Table D5).

Standard errors are computed using 1,000 bootstrap replications clustered at the regional level to ensure valid inference in this copula-based control-function setting. As a robustness check, we also report inference based on clustering at the PSU (primary sampling unit) level (see

[10] If ANC adherence affects vaccination primarily through postnatal information or knowledge channels, it should not affect birth-dose vaccines among facility-born children, whose uptake is largely determined at delivery, absent other channels.

Online Appendix E). Details on the computation of the average treatment effect are provided in Online Appendix H.

4. Results

We begin by examining the effects of antenatal care (ANC) adherence on full childhood immunization and on each vaccine-specific outcome (Section 4.1). This allows us to assess whether aggregate measures of vaccination conceal heterogeneity across vaccines. We next investigate heterogeneity by maternal education to shed light on the role of information and human capital in shaping the returns to ANC adherence (Section 4.2). We then examine whether effects vary by child sex and place of residence (Section 4.3), and we compare intended and unintended pregnancies to assess whether fertility intentions at conception shape postnatal investment responses (Section 4.4). Finally, we explore heterogeneity by ethnicity, which may proxy for persistent beliefs and trust in formal healthcare (Section 4.5).

4.1. Heterogeneity in Effect by Vaccine and Aggregation Bias

Table 2 reports the main 2sCOPE estimates, with full results presented in Online Appendix C. For comparison, we report both Probit and 2sCOPE estimates, although our discussion focuses on the latter. Panel A presents average marginal effects (AMEs) from a Probit model that treats antenatal care adherence as exogenous. Panel B reports our preferred 2sCOPE estimates, which explicitly account for endogeneity in antenatal care adherence; for this specification, we report average treatment effects (ATEs). The parameters ω_1 and ρ provide formal tests of endogeneity under the joint Gaussian assumption in the second stage, whose robustness we examine in Section 5.

When antenatal care adherence is treated as exogenous, Probit estimates indicate a positive and statistically significant association with all vaccine-specific outcomes as well as with full immunization, suggesting broadly positive effects across the vaccination schedule. Once endogeneity is accounted for, however, a markedly selective pattern emerges. Antenatal care adherence significantly increases uptake of the three-dose diphtheria–pertussis–tetanus vaccine (DTP3) by 8.2 percentage points (9.1 percent relative to the mean) and the measles vaccine by 6.3 percentage points (7.6 percent), while having no statistically significant effect on Bacillus Calmette–Guérin (BCG), Polio3, or full immunization.

This heterogeneity is mirrored in the endogeneity diagnostics. The parameters ω_1 and ρ are large and statistically significant for BCG, Polio3, and full immunization, indicating substantial positive selection into antenatal care for these outcomes. In contrast, both parameters are statistically insignificant for DTP3 and measles, suggesting that antenatal care adherence is conditionally exogenous for these vaccines.

Table 2. Effect of antenatal care adherence on childhood vaccination

	BCG	Polio3	DTP3	Measles	CV
Panel A. Probit model estimates					
ANC	0.013**	0.017*	0.042***	0.047***	0.028**
	(0.005)	(0.001)	(0.007)	(0.009)	(0.011)
Panel B. 2sCOPE model estimates					
ANC	0.024	0.022	0.082***	0.063***	0.017
	(0.015)	(0.020)	(0.016)	(0.019)	(0.022)
$\boldsymbol{\omega_1}$	0.585***	0.577***	-0.150	0.369	0.666***
	(0.162)	(0.143)	(0.145)	(0.242)	(0.154)
$\boldsymbol{\rho}$	-0.611	-0.587	0.000	-0.412	-0.658
	[-0.764; -0.362]	[-0.737; -0.347]	[-0.156; 0.143]	[-0.682; 0.029]	[-0.835; -0.352]
Sample average	0.951	0.822	0.902	0.832	0.724
N	7,660	7,660	7,660	7,660	7,660

Note: We report average marginal effects from Probit models and average treatment effects from the 2sCOPE models. CV denotes complete vaccination. All specifications include the set of demand- and supply-side controls, as well as region and survey-wave fixed effects. Standard errors are in parentheses; for the 2sCOPE models, they are based on 1,000 bootstrap replications clustered at the regional level. $\boldsymbol{\omega_1}$ denotes the coefficient on the generalized residual control function, and $\boldsymbol{\rho}$ the Gaussian copula dependence parameter (95% confidence intervals in brackets; estimates smaller than 0.001 in absolute value are rounded to 0.000). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Taken together, the results yield two main insights. The composite immunization measure exhibits aggregation bias, obscuring meaningful vaccine-specific responses. The effects of antenatal care (ANC) adherence are selective, suggesting that mechanisms beyond a purely informational channel contribute to the observed vaccination patterns. In particular, DTP3 and Polio3 are typically scheduled at the same ages and intervals and are often co-administered, yet only DTP3 responds. This is difficult to reconcile with a pure visit-bundling explanation.

A plausible interpretation is that this selectivity reflects differences between injectable and oral vaccines and the behavioral responses they elicit. DTP3 and measles are injectable, whereas Polio3 is administered orally. Oral polio vaccination is frequently reinforced through mass and door-to-door campaigns (Anya et al. 2016; Sakas et al. 2023), is easier to administer, and is less invasive (Erchick et al. 2022; Kwong et al. 2023). Injectable vaccines, by contrast, are more likely to generate anticipated discomfort and parental anxiety, including concerns about pain and mild side effects (Peretti-Watel et al. 2019; Erchick et al. 2022; Freeman et al. 2023; Kwong et al. 2023). If prior experience and confidence shape acceptance, ANC adherence may strengthen trust in providers and reduce hesitancy toward injections. During pregnancy, mothers are routinely exposed to the tetanus toxoid–containing vaccine[11] (TTCV), administered by injection (Tunçalp et al., 2017). Such exposure may increase confidence in vaccine safety and reduce anxiety about postnatal injections for the child (Peretti-Watel et al. 2019), thereby sustaining compliance with injectable childhood vaccines such as DTP3 and measles. The absence of an effect for BCG, although also injectable, is consistent with this interpretation, as BCG is typically administered at birth in health facilities, leaving limited scope for postnatal behavioral adjustment.

If antenatal counselling operates primarily through an information channel, its effects should be stronger among less-educated mothers (Barrera, 1990). We test this prediction by examining heterogeneity in the effect of antenatal care adherence across levels of maternal education.

4.2. Heterogeneous Returns by Maternal Education

We estimate the 2sCOPE model separately for three subsamples defined by maternal education. Fig. 1 shows that the positive effects of ANC adherence are concentrated among less-educated mothers. Among mothers with no formal education, ANC adherence increases DTP3 uptake by 9.1 percentage points and measles uptake by 12.8 percentage points. Among mothers with primary education, the corresponding effects are 8.4 and 15.3 percentage points. By contrast, we find no statistically significant effects among mothers with secondary or higher education. We note that point estimates for measles and the vaccination completion outcome are large and negative but imprecisely estimated[12]. Robustness check in Online Appendix E, Table E2 confirm the lack of statistical significance[13]. When pooling mothers with primary and higher education (Online Appendix D, Table D7), effects remain positive for DTP3 and measles, though the measles estimate attenuates, while estimates for other vaccination outcomes remain statistically insignificant. We view this pattern as suggestive of attenuation rather than a sharp discontinuity at higher education levels.

Taken together, our results are consistent with ANC operating in part through an informational and trust-building channel, with larger gains among mothers facing greater informational

[11] Tetanus toxoid vaccination received during ANC (by injection) has been shown to be a strong predictor of subsequent childhood immunization coverage (Nour et al., 2020; Farrenkopf et al., 2023).
[12] Although limited effective sample size among the most educated mothers cannot be ruled out, this concern is likely to be more salient for outcomes with lower prevalence (e.g., DTP3 and BCG) than for vaccination completion and measles (see Online Appendix A, Tables A8 and A10).
[13] We also report 2sCOPE estimates with 400 PSU-clustered bootstrap standard errors.

frictions and greater hesitancy toward injectable vaccines. In the next section, we examine whether responses differ by child sex and place of residence.

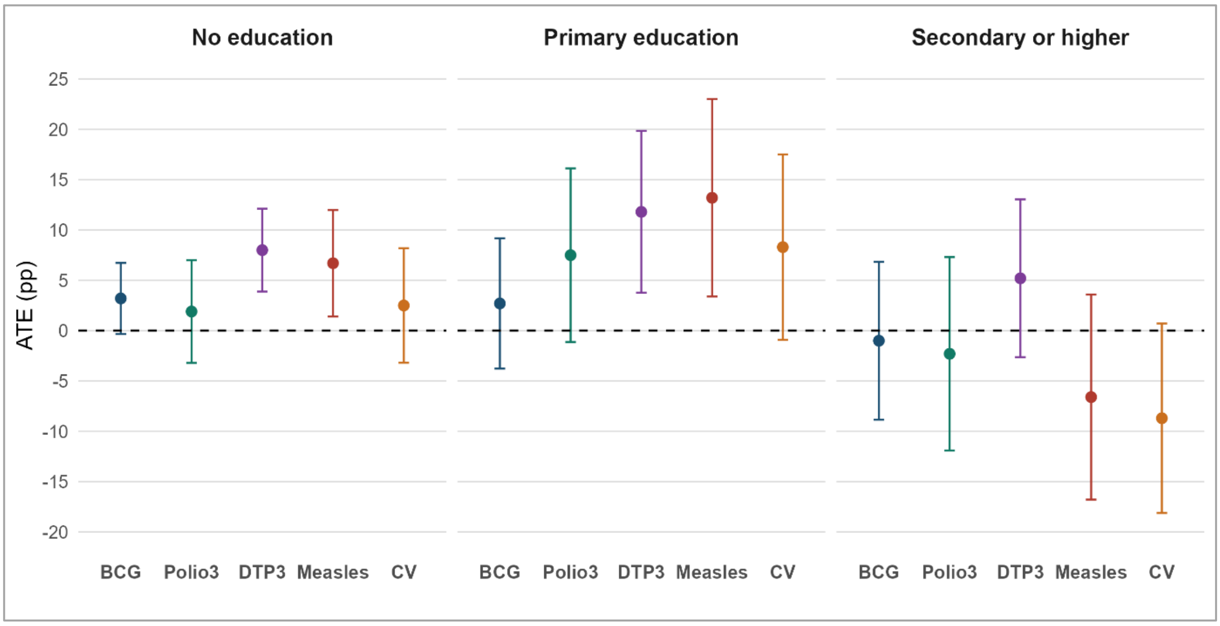

Fig.1. Heterogeneity in the effect of antenatal care adherence on childhood vaccination by maternal education level.

Note: The figure reports average treatment effects (ATEs) of antenatal care adherence on vaccination outcomes by mother's education level. Vertical bars represent 95% confidence intervals. Estimates are obtained from the 2sCOPE models and expressed as changes in outcome probabilities. All specifications include the full set of control variables.

4.3. Child Gender and the Returns to ANC Adherence

Does child sex at birth shape the returns to antenatal care adherence for postnatal vaccination outcomes? We examine this question by estimating the 2sCOPE model separately for boys and girls (Fig. 2), and then by child sex and place of residence to assess whether any gender differences vary across contexts (Fig. 3).

In the pooled sample (Fig. 2), ANC adherence increases DTP3 and measles vaccination for both boys and girls, with no evidence of differential effects by child sex. Disaggregating by place of residence suggests a more nuanced pattern. For DTP3, ANC adherence significantly increases vaccination for urban boys but not for urban girls, whereas in rural areas the effects are positive and statistically significant for both sexes, with comparable magnitudes. For measles, we find no evidence of gender differences in returns. Although the urban DTP3 estimates are suggestive of pro-son parental follow-through, the difference between urban boys and urban girls is not itself statistically significant. Following Gelman and Stern (2006), we formally test the null hypothesis of equal average treatment effects across the two groups and fail to reject it (p = 0.218; see Online Appendix F, Table F1). This warrants a cautious interpretation and underscores that differences in statistical significance across subgroups do not imply statistically significant differences between them (Gelman and Stern 2006). We therefore find no robust evidence that the marginal return to ANC adherence differs by child sex, despite suggestive point estimates in urban areas.

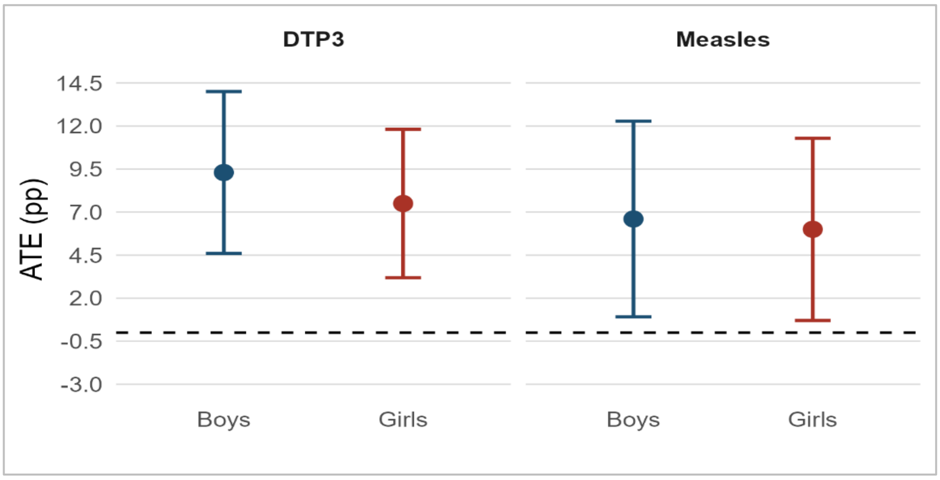

Fig. 2. Gender heterogeneity in the effect of antenatal care adherence on childhood vaccination.

Note: The figure reports average treatment effects (ATEs) of antenatal care adherence on DTP3 and measles vaccination by child gender. Vertical bars represent 95% confidence intervals. Effects are expressed in percentage points (pp). All specifications include the full set of control variables.

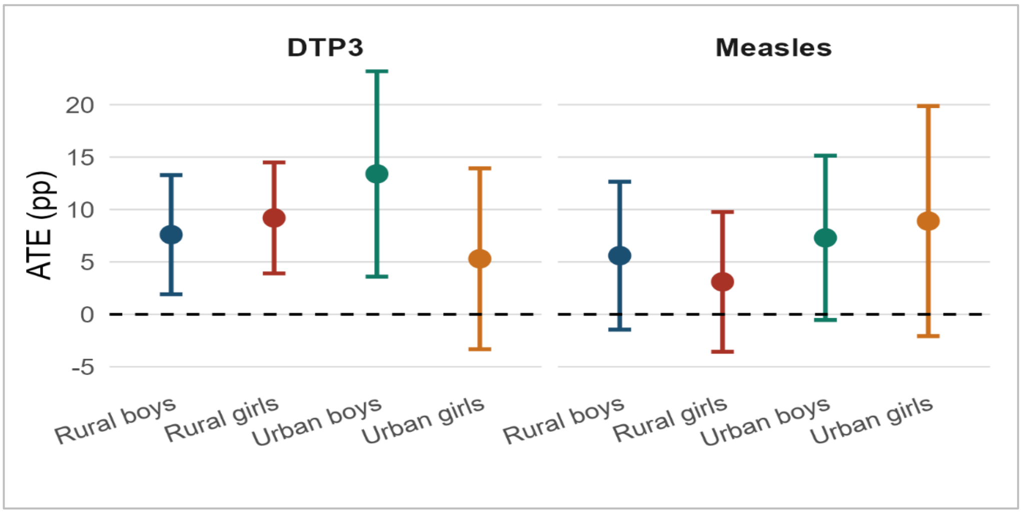

Fig.3. Gender heterogeneity in the effect of antenatal care adherence by place of residence.

Note: The figure reports average treatment effects (ATEs) of antenatal care adherence on DTP3 and measles vaccination by gender and rural–urban residence. Vertical bars represent 95% confidence intervals. Effects are expressed in percentage points (pp). All specifications include the full set of control variables.

4.4. Fertility Intention

We next investigate whether fertility intention shapes the returns to antenatal care (ANC) adherence. If the pregnancy is intended at conception, mothers may be more receptive to antenatal counselling and more likely to translate such informational inputs into postnatal vaccination effort (Joyce et al. 2000; Marston and Cleland 2003; Gipson et al. 2008). Fig. 4 provides suggestive evidence consistent with this mechanism. ANC adherence significantly increases DTP3 and measles uptake among intended births, but not among unintended births. This pattern, however, does not necessarily imply a statistically significant difference in the productivity of ANC adherence across the two groups. We formally test the null hypothesis of equal average treatment effects and fail to reject it for either outcome at the 5 percent level ($p = 0.380$ for DTP3 and $p = 0.078$ for measles; see Online Appendix F, Table F2). Our results therefore provide no robust statistical evidence that the returns to ANC adherence differ by fertility intention, although the point estimates suggest larger effects among intended births, especially for measles. We interpret the more pronounced contrast for measles as consistent with the fact that it is scheduled later in infancy (at nine months) and therefore depends more on sustained parental engagement, for which fertility intention may be a proxy.

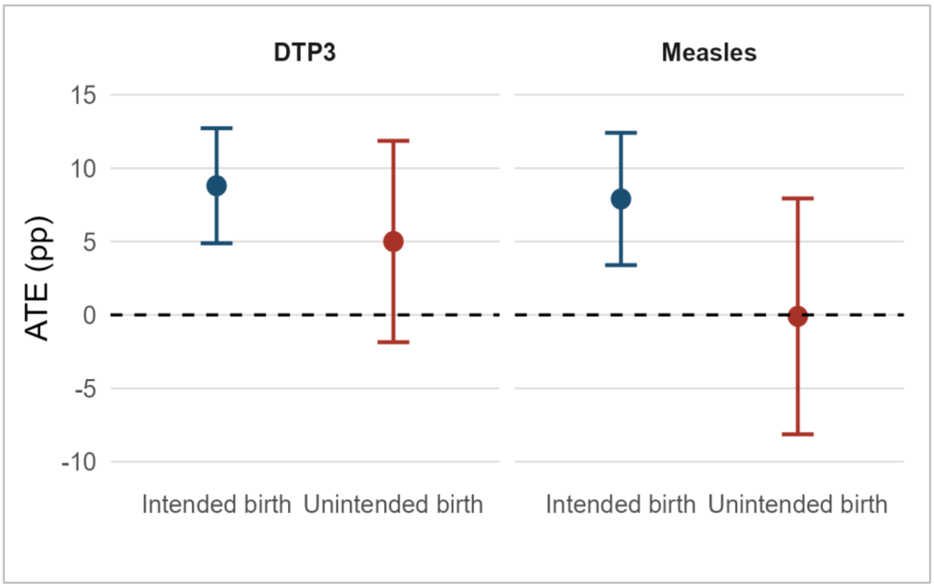

Fig.4. Heterogeneity in the effect of antenatal care adherence by fertility intention.

Note: The figure reports average treatment effects (ATEs) of antenatal care adherence on DTP3 and measles vaccination by fertility intention. Vertical bars represent 95% confidence intervals. Effects are expressed in percentage points (pp). All specifications include the full set of control variables.

4.5. Ethnicity and the Returns to ANC Adherence

We also examine whether the returns to antenatal care (ANC) adherence vary by ethnicity. Beyond a lack of information, vaccine hesitancy may also reflect misinformation or persistent beliefs. Although such attitudes are not directly observed in our data, ethnicity may proxy for differences in social norms and trust in state institutions (Zerfu et al. 2009; Arriola and Grossman 2021). To the extent that ANC exposure reinforces confidence in vaccine safety and in formal healthcare providers, its effects may vary across ethnic groups. We focus in particular on Poular/Peul households, for whom prior work documents distinct patterns of engagement with state institutions and health systems[14] (Arriola and Grossman 2021).

Table B5 in Online Appendix B reports treatment effects by ethnicity, and Table F3 in Online Appendix F tests equality across groups. Among non-Poular households, ANC adherence increases DTP3 and measles uptake by 9.4 and 8.0 percentage points, respectively. Among Poular households, the estimated effects are smaller and statistically insignificant. However, the data do not provide statistically robust evidence of ethnic heterogeneity in the returns to ANC adherence ($p = 0.154$ for DTP3 and $p = 0.206$ for measles), although point estimates are attenuated among Poular households.

5. Robustness Analysis

In this section, we conduct a series of sensitivity analyses to assess the robustness of our results (Online Appendices D–G). Overall, the findings are stable across a wide range of specifications. Alternative copula specifications in the second-stage estimation, including the Frank and Clayton copulas[15] used to model residual dependence between the unobserved determinants of antenatal care adherence and child vaccination, yield the same conclusions: no statistically significant adjusted effect on full immunization, alongside robust positive effects on DTP3 and measles uptake (Online Appendix D, Table D1).

The results are also robust to potential selection into antenatal care. Excluding high-frequency ANC visits that may proxy for complicated pregnancies, using alternative visit cutoffs (5, 6 or 7), or imposing no restriction leaves the estimates unchanged (Online Appendix D, Table D2).

[14] In Senegal, Poular/Peul are the second largest ethnic group after Wolof and are disproportionately concentrated in rural areas, with historical ties to pastoral livelihoods. Previous studies document substantially lower use of health-facility delivery and child vaccination services among Poular households (Mbengue et al., 2017; Zegeye et al., 2021), alongside localized differences in vaccination attitudes (John et al., 2021).

[15] The Frank copula is symmetric like Gaussian but flexible in both tails, while the Clayton copula is asymmetric, capturing stronger lower-tail dependence (Genest ,1987; Nelsen, 1999).

The findings are likewise stable when twin births and pregnancies with very early initiation (month 0) or very late initiation (after month 7) are included in the sample (Online Appendix D, Table D3). The heterogeneous effects by maternal education also remain unchanged (Online Appendix D, Table D6). Additional robustness checks for heterogeneity by child gender (Online Appendix D, Table D8), fertility intention (Online Appendix D, Table D9), and ethnicity (Online Appendix D, Table D10) confirm the main results. The results are also unaffected by the inclusion of additional control variables[16] that may partially capture latent heterogeneity (Online Appendix D, Table D4).

As an additional robustness check, we implement propensity score matching (PSM) (Rosenbaum and Rubin 1983; Austin 2011; Dixit et al. 2013; Tiruneh et al. 2025), which adjusts only for selection on observables (see Online Appendix G for details). Comparing mothers with similar observed characteristics but different levels of ANC utilization (fewer than four visits versus four or more visits) yields estimates that are qualitatively similar to the 2sCOPE results, although smaller in magnitude, when the propensity score is constructed using the full set of covariates in equation (2).

These robustness checks suggest that our main findings are unlikely to be driven by specific modelling assumptions or sample definitions. They also suggest that estimates that do not account for unobserved heterogeneity tend to recover positive and statistically significant effects across vaccines while understating their true magnitude.

6. Conclusion and Policy Implications

Using seven waves of the Senegal Demographic and Health Surveys (2012–2019), this paper estimates the effect of antenatal care (ANC) adherence on child vaccination outcomes using a copula-based control-function approach.

We show that treating ANC adherence as exogenous yields broadly positive associations across vaccines and full immunization. Once endogeneity is addressed, however, the pattern becomes more selective: ANC adherence significantly increases uptake only for the three-dose diphtheria-pertussis-tetanus vaccine (DTP3) and measles, by 8.2 and 6.3 percentage points, respectively. This result is economically important: aggregating across vaccines with different

[16] We additionally control for pregnancy wantedness, past pregnancy complications, women's autonomy in health-related decision-making, and age at marriage or cohabitation, as well as child sex and age.

effort costs and delivery modes masks selective gains and may mislead targeting and resource allocation.

We also contribute to the literature on heterogeneous effects of early-life investments on child health outcomes. The positive effects on DTP3 and measles are concentrated among less educated mothers, suggesting that ANC operates in part through an informational channel, with larger returns where informational frictions are greater. The fact that ANC increases uptake of DTP3 and measles, but not Polio3, further suggests that ANC may also build trust for vaccines with higher perceived risks and hesitancy, particularly injectable ones. We also find suggestive evidence of attenuated and statistically insignificant effects among urban girls for DTP3, and among unintended births and Poular/Peul households for both DTP3 and measles. However, we do not find robust evidence that the returns to ANC adherence differ across these groups, suggesting that disparities are more likely to arise through differential access to ANC or vaccination than through differential returns to ANC adherence itself.

A few limitations should be acknowledged. Vaccination and ANC data partly rely on maternal recall, which may be subject to error. However, existing evidence suggests that most DHS inaccuracies reflect interviewer or processing errors rather than systematic misreporting (Langsten and Hill, 1998). While our robustness analyses are consistent with the identifying assumption required for a causal interpretation of the 2sCOPE estimates, that assumption cannot be tested directly. We also cannot account for selection through child mortality or non-coresidence with the mother, nor can we observe facility-level quality. If vaccine counselling is inconsistently delivered across facilities (Návar et al., 2007), our estimates may understate the information channel, although the pattern of results makes this less likely.

The findings have direct policy implications. Strengthening ANC counselling and improving adherence during pregnancy, particularly among less educated women, may reduce information gaps and improve postnatal follow-through. Improving mothers' experience with injectable vaccines during pregnancy, such as tetanus vaccination, may also build confidence in vaccine safety and increase uptake of injectable childhood vaccines. More broadly, given limited evidence of heterogeneity in returns, policy should prioritize reducing access barriers to ANC and immunization services. Future work should prioritize vaccine-specific outcomes rather than composite immunization measures.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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ONLINE APPENDIX FOR "HETEROGENEOUS RETURNS TO ANTENATAL CARE: EVIDENCE FROM CHILDHOOD IMMUNIZATION IN SENEGAL".

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Online Appendix A. Data Processing and Descriptive Statistics

Table A1. DHS sample sizes before and after data processing

Data processing	DHS Waves						
	2012 - 2013	2015	2016	2017	2018	2019	Total
Initial sample (children aged 0–59 months)	6862	6935	6725	12185	6719	6125	45551
Study population (children aged 12–23 months)	1329	1310	1323	2390	1337	1183	8872
One child per mother (twin case adjustment)	1308	1291	1304	2357	1312	1167	8739
Final analytic sample	1149	1167	1144	2115	1183	1058	7816
Full sample	1117	1141	1110	2087	1162	1043	7660

Note: This table reports sample sizes by DHS wave at each step of sample construction. The initial sample includes children aged 0–59 months; the study population includes children aged 12–23 months. We retain one child per mother; in multiple births, we keep the firstborn to avoid duplicate maternal observations. The final analytic sample reflects all data-processing steps[18], whereas the full sample includes observations remaining after applying ANC-related restrictions[19] (see Section 2.1 for details and Appendix C for robustness checks).

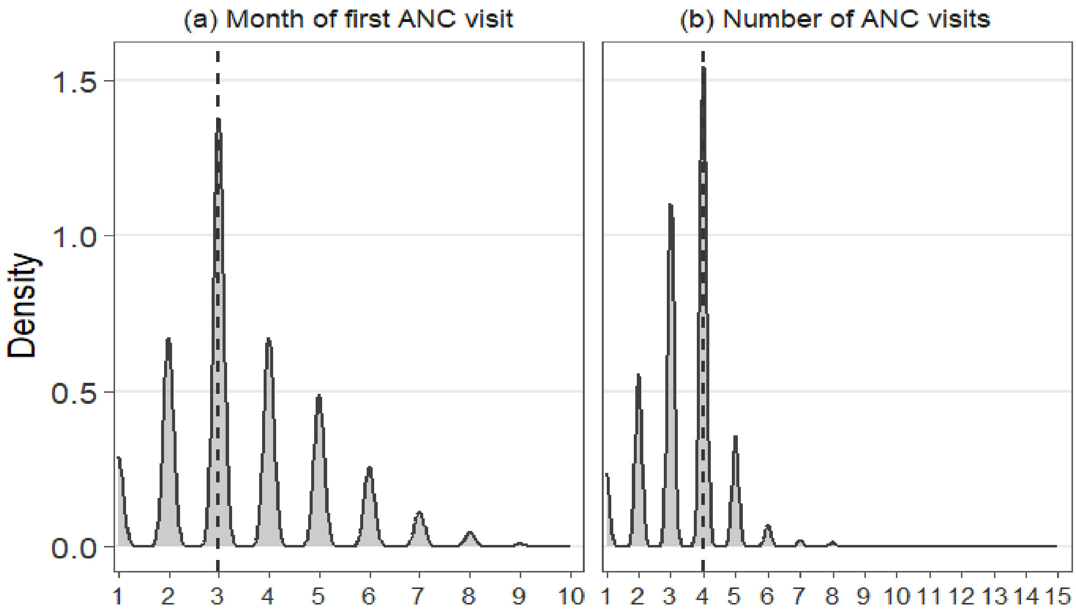

Fig. A1. Distribution of ANC initiation month and number of visits

Note: This figure displays the distributions of (a) the month of the first antenatal care (ANC) visit and (b) the total number of ANC visits during pregnancy, prior to imposing any sample restrictions. Vertical dashed lines indicate the WHO-recommended thresholds.

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[18] These steps include dropping observations with missing values and DHS special codes (e.g., 98 "don't know") in key variables, including ANC utilization measures.

[19] From the final analytic sample of 7,816 observations, imposing the ANC-visit restriction drops 13 observations (ANC visits ≥ 9). Imposing the ANC-initiation restriction drops 143 observations, including 32 with initiation month 0 and 111 with initiation month 8–10 (91, 19, and 1, respectively). These restrictions reduce the sample from 7,816 to 7,660.

Table A2. Routine childhood vaccination schedule in Senegal (EPI/WHO)

Vaccine name	Disease	Age of 1st dose	Number of doses required	Interval 1st to 2nd dose	Interval 2nd to 3rd dose
BCG	Tuberculosis	At birth	1		
Polio (OPV)	Poliomyelitis	6 weeks	3	4 weeks	4 weeks
DTP (Pentavalent)	Diphtheria, Tetanus, Pertussis, Hepatitis B, Haemophilus influenzae type b	6 weeks	3	4 weeks	4 weeks
Measles–Rubella (MR)	Measles	9 months	1		

Note: This table summarizes the routine immunization schedule recommended by the World Health Organization (WHO) under the Expanded Programme on Immunization (EPI) and implemented by Senegal's Ministry of Health. Ages refer to the recommended timing of the first dose; intervals indicate the recommended spacing between subsequent doses. The polio birth dose (Polio 0) and booster doses are not included in the study's standard indicator of full basic immunization. Source: Authors' calculations based on WHO/EPI recommendations.

Fig. A2 and Table A3 report national and regional full vaccination coverage among children aged 12–23 months in Senegal using DHS waves from 2012–2019.

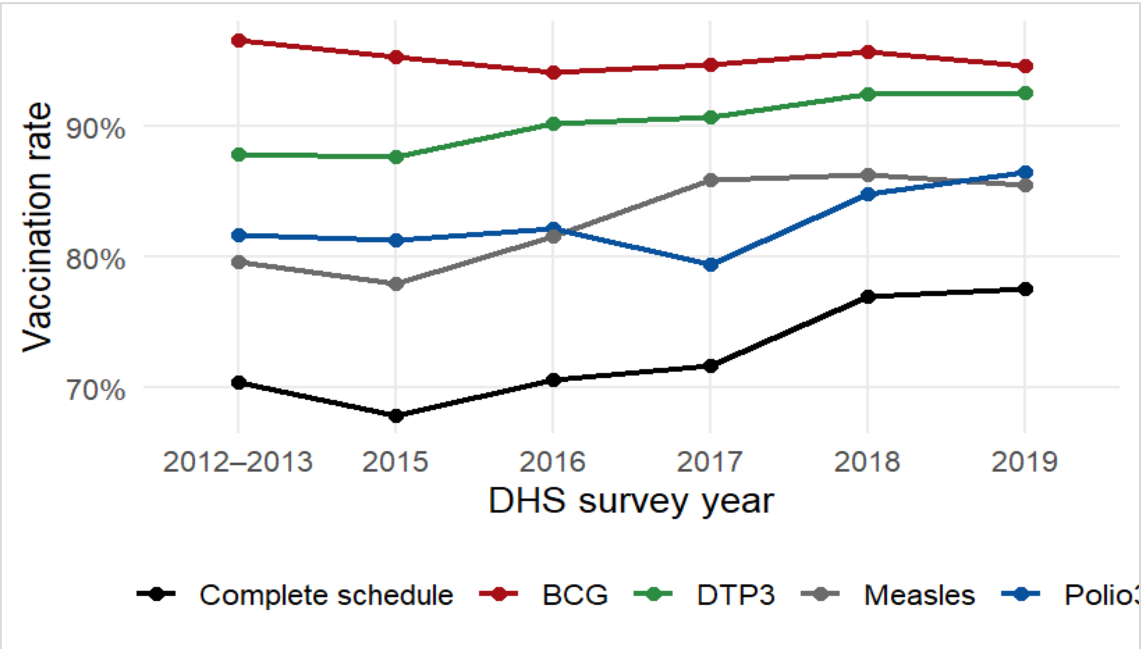

Fig. A2. Trends in vaccination coverage by vaccines and DHS survey year

Note: The figure plots vaccination rates (percent) across DHS survey years. Complete schedule denotes receipt of all vaccines in the standard schedule as defined in the paper.

National coverage increases over time, but large and persistent regional disparities remain in every round. These patterns are consistent because the national series is a population-weighted average: improvements in some regions can offset stagnation or declines elsewhere.

Accordingly, the modest national change between 2018 and 2019 reflects offsetting regional movements, plausibly related to heterogeneity in local health-system capacity and geographic access. Overall, recent national gains appear driven by uneven regional progress rather than broad-based convergence.

Table A3. Temporal trends and regional disparities in full vaccination coverage among children aged 12–23 months, Senegal (2012–2019)

	Years of the DHS waves					
	2012/2013	2015	2016	2017	2018	2019
	National level (%)					
SENEGAL	70.5	67.9	70.6	71.7	77.0	77.6
	Regional levels (%)					
Dakar	72.9	79.1	77.8	86.1	81.3	75.4
Ziguinchor	73.8	62.5	89.2	81.3	93.3	82.1
Diourbel	65.4	66.7	72.9	66.7	72.3	78.1
Saint-Louis	82.7	82.0	73.3	82.8	70.8	87.7
Tambacounda	48.8	46.9	40.8	49.0	73.2	77.6
Kaolack	77.7	73.9	68.5	77.7	74.3	82.5
Thiès	88.0	82.2	88.5	82.5	85.4	77.7
Louga	67.6	64.4	60.3	74.1	71.8	76.1
Fatick	81.2	73.3	85.7	75.7	92.9	81.5
Kolda	51.1	67.0	58.5	61.4	58.6	63.0
Matam	74.4	57.6	71.0	67.5	67.9	77.3
Kaffrine	67.0	72.6	77.4	77.9	92.3	82.1
Kédoudou	67.9	55.0	56.9	38.2	55.1	80.0
Sédhiou	69.1	63.2	73.9	73.9	79.5	68.4

Trends and Regional Disparities by Vaccine

Table A4. Temporal trends and regional disparities in BCG vaccination among children aged 12–23 months in Senegal (2012–2019)

	Years of the DHS waves					
	2012/2013	2015	2016	2017	2018	2019
	National level (%)					
SENEGAL	96.6	95.3	94.1	94.7	95.7	94.6
	Regional levels (%)					
Dakar	100.0	100.0	98.1	100.0	98.7	98.2
Ziguinchor	100.0	100.0	97.3	99.1	97.8	100.0
Diourbel	98.1	98.0	95.8	95.0	91.1	97.9
Saint-Louis	98.7	100.0	94.7	100.0	94.4	98.2
Tambacounda	82.5	81.5	77.6	81.5	95.1	89.6
Kaolack	99.1	100.0	99.1	100.0	97.2	95.8
Thiès	100.0	100.0	100.0	100.0	98.9	96.8
Louga	98.6	95.4	94.5	95.8	98.7	95.8
Fatick	100.0	98.8	98.9	98.2	100.0	94.4
Kolda	93.5	96.5	93.6	89.0	90.8	87.7
Matam	96.3	89.4	92.8	94.4	97.4	98.9
Kaffrine	93.2	91.6	96.2	98.4	98.5	91.6
Kédoudou	94.6	83.3	80.6	74.5	79.6	93.3
Sédhiou	98.8	96.1	95.7	94.3	95.2	88.6

Tables A4–A7 report national and regional vaccination coverage among children aged 12–23 months in Senegal using DHS waves from 2012 to 2019. Estimates are disaggregated by vaccine and survey wave to highlight time trends and geographic disparities.

Table A5. Temporal trends and regional disparities in Polio3 vaccination among children aged 12–23 months in Senegal (2012–2019)

	Years of the DHS waves					
	2012/2013	2015	2016	2017	2018	2019
	National level (%)					
SENEGAL	81.6	81.2	82.2	79.4	84.8	86.5
	Regional levels (%)					
Dakar	83.1	90.7	85.2	88.1	92.0	84.2
Ziguinchor	78.6	75.0	89.2	83.0	97.8	84.6
Diourbel	81.3	80.8	86.5	75.5	83.2	84.4
Saint-Louis	88.0	90.0	85.3	86.1	75.0	91.2
Tambacounda	56.3	63.0	63.2	63.6	81.7	86.6
Kaolack	92.0	85.8	81.1	85.4	83.5	86.7
Thiès	99.0	86.1	89.7	89.3	94.4	86.2
Louga	83.8	81.6	79.5	83.9	78.2	84.5
Fatick	88.4	81.4	87.9	82.8	95.2	87.0
Kolda	65.2	82.6	72.3	70.3	74.7	76.5
Matam	81.7	81.8	88.4	78.1	76.9	90.9
Kaffrine	81.8	87.4	86.8	83.7	93.1	91.6
Kédoudou	80.4	70.0	70.8	55.9	65.3	91.1
Sédhiou	77.8	77.6	87.0	79.0	88.0	86.1

Table A6. Temporal trends and regional disparities in DTP3 vaccination among children aged 12–23 months in Senegal (2012–2019)

	Years of the DHS waves					
	2012/2013	2015	2016	2017	2018	2019
	National level (%)					
SENEGAL	87.8	87.6	90.2	90.7	92.4	92.5
	Regional levels (%)					
Dakar	93.2	100.0	96.3	99.3	97.3	93.0
Ziguinchor	95.2	95.8	94.6	94.6	97.8	97.4
Diourbel	81.3	89.9	93.8	89.3	90.1	92.7
Saint-Louis	94.7	98.0	90.7	97.5	91.7	94.7
Tambacounda	68.8	60.5	76.3	74.8	86.6	91.0
Kaolack	92.9	93.3	92.8	91.5	91.7	93.3
Thiès	99.0	96.0	97.7	98.9	98.9	95.7
Louga	89.2	85.1	80.8	90.9	93.6	94.4
Fatick	94.2	91.9	98.9	94.1	100.0	90.7
Kolda	79.3	86.1	80.9	87.6	80.5	80.2
Matam	87.8	80.3	92.8	88.8	92.3	100.0
Kaffrine	81.8	89.5	94.3	94.2	97.7	91.6
Kédoudou	87.5	75.0	80.6	70.6	77.6	91.1
Sédhiou	90.1	88.2	91.3	90.3	92.8	89.9

Table A7. Temporal trends and regional disparities in Measles vaccination among children aged 12–23 months in Senegal (2012–2019)

	Years of the DHS waves					
	2012/2013	2015	2016	2017	2018	2019
	National level (%)					
SENEGAL	79.6	77.9	81.6	85.9	86.3	85.5
	Regional levels (%)					
Dakar	81.4	88.4	88.9	96.7	86.7	82.5
Ziguinchor	90.5	83.3	94.6	92.9	93.3	97.4
Diourbel	68.2	79.8	79.2	84.9	82.2	85.4
Saint-Louis	90.7	90.0	81.3	96.7	88.9	91.2
Tambacounda	63.8	51.9	60.5	68.2	81.7	88.1
Kaolack	83.0	82.8	82.9	86.9	83.5	90.8
Thiès	88.0	92.1	97.7	91.0	91.0	88.3
Louga	77.0	77.0	65.8	88.1	85.9	84.5
Fatick	87.0	87.2	95.6	88.8	96.4	90.7
Kolda	71.7	73.9	69.1	84.1	70.1	72.8
Matam	85.4	62.1	79.7	79.4	85.9	85.2
Kaffrine	72.7	82.1	89.6	89.5	97.7	84.2
Kédoudou	76.8	66.7	76.4	55.9	73.5	86.7
Sédhiou	86.4	72.3	84.1	90.9	85.5	75.9

Descriptive Statistics on Control Variables

Table A8. Descriptive statistics on qualitative control variables

Variable	Categories	Proportion	Full vaccination	ANC Adherence
Place of residence	rural	0.698	0.717	0.470
	urban	0.302	0.740	0.645
Birth order	1st birth	0.229	0.758	0.601
	2nd birth	0.183	0.717	0.535
	3rd birth and more	0.588	0.713	0.489
Media exposure	access to information	0.893	0.737	0.539
	no access	0.107	0.617	0.389
Mother's education	no education	0.645	0.695	0.471
	primary	0.209	0.754	0.576
	secondary and higher	0.146	0.812	0.675
Ethnicity	wolof	0.338	0.789	0.539
	poular	0.324	0.716	0.550
	others	0.339	0.668	0.481
Relationship to household head	wife	0.407	0.686	0.471
	daughter and in-law	0.345	0.778	0.548
	others	0.248	0.724	0.552
Mother’s occupational category	not employed	0.410	0.721	0.552
	sales	0.192	0.724	0.589
	others	0.398	0.728	0.461
Partner’s occupational category	not employed	0.059	0.732	0.514
	civil servant	0.084	0.801	0.696
	sales	0.164	0.724	0.546
	farmer	0.280	0.688	0.404
	others	0.412	0.732	0.561
Polygamy	yes	0.666	0.735	0.538
	no	0.334	0.703	0.494

Note: ANC adherence refers to the completion of the four recommended antenatal care visits, as defined by the WHO.

Table A9. Descriptive statistics on quantitative control variables

Variable	Min	Max	Mean	SD
Number of children	0	24	3.572	2.430
Mother's age	15	49	28.570	6.855
Household wealth index	-20.154	31.294	-0.881	6.257
Number of hospitals*	0	0.463	0.194	0.097
Number of health centers*	0.305	2.077	0.734	0.346
Number of health posts*	3.322	21.048	11.041	4.001

Note: The number of children refers to those under age 5 in the household. * Counts per 100,000 inhabitants at the regional level.

Table A10. Descriptive statistics for control qualitative variables by vaccine

Variable	Categories	Vaccines			
		BCG	Polio3	DTP3	Measles
Place of residence	rural	0.939	0.822	0.893	0.823
	urban	0.980	0.822	0.926	0.852
Birth order	1st birth	0.964	0.842	0.928	0.866
	2nd birth	0.946	0.816	0.901	0.834
	3rd birth and more	0.948	0.816	0.893	0.818
Media exposure	access to information	0.960	0.831	0.913	0.843
	no access	0.877	0.746	0.816	0.735
Mother's education	no education	0.940	0.808	0.880	0.800
	primary	0.967	0.837	0.924	0.863
	secondary and higher	0.982	0.863	0.971	0.925
Ethnicity	wolof	0.930	0.867	0.934	0.877
	poular	0.948	0.806	0.900	0.830
	others	0.976	0.792	0.873	0.788
Relationship to household head	wife	0.938	0.801	0.874	0.795
	daughter and in-law	0.962	0.853	0.924	0.879
	others	0.956	0.820	0.913	0.834
Mother's occupational category	not employed	0.958	0.817	0.902	0.830
	sales	0.961	0.817	0.916	0.836
	others	0.940	0.829	0.896	0.832
Partner's occupational category	not employed	0.965	0.809	0.930	0.857
	civil servant	0.988	0.860	0.961	0.919
	sales	0.965	0.819	0.903	0.840
	farmer	0.921	0.812	0.876	0.799
	others	0.957	0.824	0.905	0.830
Polygamy	yes	0.957	0.832	0.912	0.843
	no	0.941	0.801	0.884	0.809

Note: BCG is the single dose of the Bacillus Calmette–Guérin vaccine against tuberculosis; Polio3 refers to the three doses of the polio vaccine; DTP3 refers to the three doses of the diphtheria–tetanus–pertussis (DTP) vaccine; and Measles refers to the first dose of the measles vaccine.

Online Appendix B. First-Stage Predictive Content and Separability Checks

This appendix reports two additional checks that support the conceptual framework. First, WHO-recommended first-trimester initiation (up to 12 weeks of gestation), which captures a calendar-feasibility constraint, explains a substantial share of the variation in adherence to the WHO-recommended four-visit schedule and materially improves predictive fit (Table B1).

Second, initiation and adherence may not be separately identified when jointly included in the vaccination equation (Table B2). Because initiation and adherence are closely related margins of ANC utilization (Tunçalp et al. 2017; Jiwani et al. 2020; Asmare et al. 2025), the corresponding coefficients are unstable and are therefore not interpreted as independent effects. Accordingly, we focus on adherence in the vaccination equation and treat initiation primarily as a feasibility predictor in the adherence equation.

Table B1. Predictive content of initiation timing for ANC adherence

	(1)	(2)	(3)	(4)	(5)
WHO first-trimester initiation	0.455***	0.445***	—	—	0.418***
	(0.010)	(0.010)			(0.010)
Adj. R^2	0.198	0.213	0.066	0.075	0.234
Partial R^2	0.198	0.192	—	—	0.172
Brier score (MSE)	0.200	0.196	0.232	0.223	0.190
Demand-side factors	no	no	yes	yes	yes
Supply-side factors	no	yes	no	yes	yes
Regions and waves fixed effects	no	yes	no	yes	yes
N	7,660	7,660	7,660	7,660	7,660

Note: This table reports linear probability models for ANC adherence. Columns (1)–(5) progressively add demand-side factors, supply-side factors, and region and survey-wave fixed effects, as indicated. Demand-side factors include mother's age, education, ethnicity, occupational status, and media exposure; child birth order; partner occupational status; and household characteristics (place of residence, wealth index, mother's relationship to the household head, number of children, and polygamy). Supply-side factors include the number of hospitals, health centers, and health posts per 100,000 inhabitants. Partial R^2 is the incremental explanatory power of first-trimester initiation. The Brier score is the mean squared error of predicted adherence probabilities. Standard errors are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table B2. Checks on the Separability of Initiation and Adherence Effects

	BCG	Polio3	DPT3	Measles	CV
ANC	-1.248**	0.450	0.402	0.530	0.049
	(0.595)	(0.634)	(0.618)	(0.425)	(0.581)
W	0.612**	-0.157	0.053	-0.112	0.003
	(0.299)	(0.256)	(0.263)	(0.169)	(0.222)
All controls	yes	yes	yes	yes	yes
N	7,660	7,660	7,660	7,660	7,660

Note: This table reports 2sCOPE second-stage coefficients from specifications that jointly include ANC adherence and first-trimester initiation (W) in the vaccination equation as a separability check. Because initiation and adherence are closely related margins of ANC utilization, these coefficients are unstable. All specifications include the full set of control variables. Standard errors (in parentheses) are based on 1,000 bootstrap replications clustered at the regional level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Online Appendix C. Main Results with all Control Variables

Table C1. Main results with full set of control variables

Variable	Categories	BCG		Polio3		DTP3		Measles		CV	
		2sCOPE	Probit	2sCOPE	Probit	2sCOPE	Probit	2sCOPE	Probit	2sCOPE	Probit
ANC		0.212	0.147***	0.080	0.067*	0.527***	0.275***	0.267***	0.203***	0.056	0.089***
		(0.163)	(0.061)	(0.124)	(0.039)	(0.119)	(0.048)	(0.086)	(0.040)	(0.121)	(0.033)
Mother’s age		0.014***	0.015***	0.007*	0.007**	0.012***	0.012***	0.012***	0.012***	0.011***	0.012***
		(0.005)	(0.005)	(0.003)	(0.003)	(0.004)	(0.004)	(0.004)	(0.004)	(0.003)	(0.003)
Place of residence	rural (ref)										
	urban	0.255***	0.273***	-0.108**	-0.121***	-0.113**	-0.105*	-0.098*	-0.102**	-0.081*	-0.095**
		(0.068)	(0.084)	(0.045)	(0.045)	(0.048)	(0.058)	(0.059)	(0.048)	(0.046)	(0.042)
Birth order	1st birth (ref)										
	2nd birth	-0.123	-0.143	-0.098***	-0.108*	-0.113*	-0.123*	-0.095	-0.102*	-0.113**	-0.126**
		(0.092)	(0.092)	(0.034)	(0.056)	(0.059)	(0.072)	(0.070)	(0.060)	(0.051)	(0.052)
	3rd birth and more	-0.114*	-0.127	-0.080	-0.088	-0.117*	-0.130*	-0.112	-0.120**	-0.102*	-0.114**
		(0.065)	(0.095)	(0.053)	(0.058)	(0.062)	(0.073)	(0.072)	(0.061)	(0.054)	(0.053)
Media exposure	access to information (ref)										
	no access	-0.182***	-0.215***	-0.114**	-0.131**	-0.147***	-0.163***	-0.131**	-0.143**	-0.113**	-0.133**
		(0.060)	(0.071)	(0.051)	(0.055)	(0.038)	(0.061)	(0.054)	(0.055)	(0.050)	(0.051)
Mother's education	no education (ref)										
	primary	0.183***	0.205***	0.099**	0.110**	0.169**	0.178***	0.195***	0.208***	0.146***	0.167***
		(0.067)	(0.078)	(0.050)	(0.046)	(0.072)	(0.058)	(0.043)	(0.048)	(0.041)	(0.042)
	secondary and higher	0.341***	0.372***	0.198***	0.206***	0.571***	0.591***	0.480***	0.503***	0.305***	0.332***
		(0.087)	(0.118)	(0.065)	(0.061)	(0.105)	(0.094)	(0.065)	(0.070)	(0.053)	(0.056)
Household wealth index		0.018**	0.021***	0.007	0.008**	0.016***	0.017***	0.005	0.005	0.004	0.004
		(0.007)	(0.007)	(0.005)	(0.003)	(0.006)	(0.004)	(0.004)	(0.004)	(0.005)	(0.003)
Number of children		-0.007	-0.007		0.018**	0.004	0.004	0.010	0.010	0.019***	0.022***
		(0.006)	(0.011)		(0.008)	(0.011)	(0.009)	(0.008)	(0.008)	(0.005)	(0.006)
Ethnicity	others (ref)										
	wolof	-0.136	-0.142	-0.102*	-0.111**	-0.101*	-0.100	-0.197***	-0.205***	-0.170***	-0.192***
		(0.109)	(0.087)	(0.059)	(0.051)	(0.057)	(0.064)	(0.046)	(0.053)	(0.045)	(0.047)
	poular	-0.199*	-0.212**	-0.172***	-0.188***	-0.197**	-0.193***	-0.253***	-0.264***	-0.206***	-0.233***
		(0.113)	(0.088)	(0.059)	(0.051)	(0.079)	(0.065)	(0.057)	(0.053)	(0.044)	(0.047)
Relationship to household head	others (ref)										
	wife	-0.030	-0.038	-0.007	-0.010	-0.098*	-0.105**	-0.041	-0.046	-0.021	-0.026
		(0.053)	(0.069)	(0.047)	(0.043)	(0.058)	(0.053)	(0.048)	(0.044)	(0.040)	(0.040)
	daughter and in-law	0.102**	0.109	0.093**	0.101**	0.065	0.064	0.203***	0.211***	0.146***	0.164***
		(0.049)	(0.077)	(0.040)	(0.047)	(0.058)	(0.059)	(0.043)	(0.050)	(0.037)	(0.043)

Mother's occupational category	not employed (ref)										
	sales	-0.170***	-0.193**	-0.010	-0.011	0.045	0.046	-0.013	-0.014	-0.025	-0.028
		(0.057)	(0.084)	(0.038)	(0.050)	(0.040)	(0.064)	(0.044)	(0.052)	(0.040)	(0.046)
	others	0.072	0.078	0.116*	0.127***	0.165***	0.159***	0.127***	0.131***	0.097*	0.110***
		(0.063)	(0.068)	(0.068)	(0.043)	(0.063)	(0.053)	(0.041)	(0.045)	(0.053)	(0.039)
Partner's occupational category	not employed (ref)										
	civil servant	0.303**	0.354*	0.086	0.082	0.177**	0.202	0.265***	0.278**	0.148*	0.151
		(0.136)	(0.207)	(0.094)	(0.099)	(0.078)	(0.142)	(0.091)	(0.112)	(0.082)	(0.092)
	sales	-0.018	0.000	-0.036	-0.044	-0.103	-0.089	0.027	0.032	-0.000	-0.007
		(0.138)	(0.152)	(0.069)	(0.089)	(0.080)	(0.116)	(0.069)	(0.095)	(0.057)	(0.082)
	farmer	-0.100	-0.106	0.059	0.060	0.005	0.009	0.019	0.019	0.020	0.017
		(0.125)	(0.143)	(0.059)	(0.087)	(0.068)	(0.113)	(0.079)	(0.092)	(0.059)	(0.080)
	others	-0.064	-0.056	0.005	0.003	-0.086	-0.073	-0.023	-0.021	0.030	0.029
		(0.128)	(0.138)	(0.076)	(0.081)	(0.083)	(0.107)	(0.054)	(0.087)	(0.071)	(0.075)
Polygamy	no (ref)										
	yes	0.059	0.064	0.079**	0.087**	0.100*	0.100**	0.076**	0.079*	0.044	0.050
		(0.052)	(0.061)	(0.040)	(0.040)	(0.054)	(0.048)	(0.037)	(0.041)	(0.037)	(0.037)
Number of hospitals*		1.397	1.552	0.255	0.299	0.398	0.352	-1.047	-1.117	-0.520	-0.583
		(1.075)	(1.912)	(0.620)	(0.997)	(0.740)	(1.222)	(1.451)	(1.020)	(0.830)	(0.908)
Number of health centers*		0.964**	1.073***	0.572	0.643***	0.790**	0.781***	0.812**	0.850***	0.679*	0.783***
		(0.481)	(0.326)	(0.412)	(0.234)	(0.378)	(0.271)	(0.358)	(0.239)	(0.378)	(0.217)
Number of health posts*		-0.006	-0.008	0.041***	0.046***	-0.003	-0.006	0.012	0.012	0.032	0.037**
		(0.014)	(0.025)	(0.014)	(0.016)	(0.014)	(0.020)	(0.022)	(0.017)	(0.019)	(0.015)
Constant		0.665	0.766	0.165	0.177	0.462	0.535	0.370	0.412	-0.007	-0.024
		(0.600)	(0.922)	(0.407)	(0.502)	(0.500)	(0.617)	(0.763)	(0.514)	(0.473)	(0.458)
ω_1		0.585***		0.577***		-0.150		0.369		0.666***	
		(0.162)		(0.143)		(0.145)		(0.242)		(0.154)	
ρ		-0.611		-0.587		-7.2e-05		-0.412		-0.658	
		[-0.764; -0.362]		[-0.737; -0.347]		[-0.156; 0.143]		[-0.682; 0.029]		[-0.835; -0.352]	
N		7,660	7,660	7,660	7,660	7,660	7,660	7,660	7,660	7,660	7,660

Note: This table reports estimates of the structural coefficient γ capturing the effect of antenatal care adherence on vaccination outcomes, from the 2sCOPE and Probit models, along with coefficient estimates for the full set of control variables. CV denotes complete vaccination. ω_1 denotes the coefficient on the generalized residual control function, and ρ the Gaussian copula dependence parameter (95% confidence intervals in brackets; estimates smaller than 0.001 in absolute value are rounded to 0.000). *Counts per 100,000 inhabitants at the regional level. Standard errors (in parentheses) are bootstrapped with 1,000 replications at the regional level. *** p<0.01, ** p<0.05, * p<0.1.

Online Appendix D. Robustness Checks

Table D1. Robustness of main results to alternative copula specifications

	BCG	Polio3	DTP3	Measles	CV
Panel A. 2sCOPE model with Frank Copula					
ANC	0.196	0.076	0.514***	0.273***	0.076
	(0.142)	(0.085)	(0.107)	(0.090)	(0.074)
Panel B. 2sCOPE model with Clayton Copula					
ANC	0.269*	0.105	0.527***	0.286***	0.097
	(0.148)	(0.111)	(0.104)	(0.091)	(0.080)
Sample average	0.951	0.822	0.902	0.832	0.724
N	7,660	7,660	7,660	7,660	7,660

Note: This table reports robustness checks using alternative copula functions (Frank and Clayton) in the second-stage joint estimation of the 2sCOPE model. We report the structural coefficient $\boldsymbol{\gamma}$ rather than average treatment effects. All specifications include the full set of control variables. The Frank copula is symmetric, like the Gaussian copula, but allows for flexible tail dependence, whereas the Clayton copula is asymmetric and captures stronger lower-tail dependence (Nelsen, 1999; Genest, 1987). Standard errors, reported in parentheses, are based on 1,000 bootstrap replications clustered at the regional level. Full results are available upon request. *** p < 0.01, ** p < 0.05, * p < 0.10.

Table D2. Sensitivity analysis on alternative cutoffs of ANC visits

	ANC visits restrictions			
	No restriction	<=7	<=6	<=5
Panel A. Complete vaccination (CV)				
ANC	0.059	0.060	0.063	0.073
	(0.079)	(0.079)	(0.082)	(0.080)
Panel B. BCG vaccination				
ANC	0.212	0.212	0.231*	0.234*
	(0.138)	(0.138)	(0.135)	(0.139)
Panel C. Polio3 vaccination				
ANC	0.081	0.087	0.086	0.089
	(0.083)	(0.079)	(0.085)	(0.088)
Panel D. DTP3 vaccination				
ANC	0.528***	0.528***	0.532***	0.541***
	(0.106)	(0.101)	(0.103)	(0.106)
Panel E. Measles vaccination				
ANC	0.271***	0.263***	0.263***	0.263***
	(0.088)	(0.091)	(0.091)	(0.091)
N	7,670	7,632	7,592	7,452

Note: This table reports robustness checks for potential selection related to antenatal care utilization by imposing alternative cutoffs on the total number of ANC visits (e.g., 7, 6, or 5), and by imposing no upper restriction on ANC visits. Estimates are stable across columns. We report the structural coefficient $\boldsymbol{\gamma}$ rather than average treatment effects. All specifications include the full set of control variables. Standard errors, reported in parentheses, are based on 1,000 bootstrap replications clustered at the regional level. Full results are available upon request. *** p < 0.01, ** p < 0.05, * p < 0.10.

Table D3. Sensitivity to ANC initiation and twins restrictions

	ANC initiation restriction		*All twins included*
	No restriction	<=7	
Panel A. Complete vaccination (CV)			
ANC	0.078	0.056	0.056
	(0.118)	(0.122)	(0.122)
Panel B. BCG vaccination			
ANC	0.207	0.204	0.210
	(0.152)	(0.135)	(0.164)
Panel C. Polio3 Vaccination			
ANC	0.095	0.080	0.075
	(0.121)	(0.085)	(0.125)
Panel D. DTP3 Vaccination			
ANC	0.551***	0.520***	0.512***
	(0.098)	(0.104)	(0.116)
Panel E. Measles Vaccination			
ANC	0.302***	0.271***	0.271***
	(0.086)	(0.085)	(0.084)
N	7,803	7,771	7,782

Note: This table reports 2sCOPE estimates of the effect of ANC adherence on child vaccination outcomes under alternative restrictions on the month of ANC initiation (no restriction; initiation $\leq$ 7 months) and under a sample definition that includes all twin births (rather than retaining one child per mother). Estimates are stable across columns. We report the structural coefficient $\boldsymbol{\gamma}$ rather than average treatment effects. All specifications include the full set of control variables. Standard errors (in parentheses) are bootstrapped with 1,000 replications at the regional level. Full results available upon request. *** p<0.01, ** p<0.05, * p<0.1.

Table D4. Sensitivity to Additional Control Variables

	(a) main specification	(a) + wantedness of the child	(a) + past pregnancy complications	(a) + mother's heath decision autonomy	(a) + age at marriage/cohabitation	(a) + age and sex of the child	(a) + all additional controls
Panel A. Full vaccination							
ANC	0.056	0.060	0.054	0.062	0.062	0.055	0.062
	(0.121)	(0.125)	(0.123)	(0.125)	(0.120)	(0.121)	(0.132)
Panel B. BCG vaccination							
ANC	0.212	0.218	0.215	0.238	0.220	0.205	0.239
	(0.163)	(0.170)	(0.165)	(0.167)	(0.164)	(0.162)	(0.175)
Panel C. Polio3 Vaccination							
ANC	0.080	0.077	0.076	0.080	0.078	0.089	0.084
	(0.124)	(0.128)	(0.124)	(0.130)	(0.122)	(0.123)	(0.131)
Panel D. DPT3 Vaccination							
ANC	0.527***	0.518***	0.525***	0.523***	0.522***	0.520***	0.506***
	(0.119)	(0.125)	(0.121)	(0.129)	(0.124)	(0.120)	(0.134)
Panel E. Measles Vaccination							
ANC	0.267***	0.280***	0.265***	0.282***	0.277***	0.250***	0.274***
	(0.086)	(0.092)	(0.087)	(0.099)	(0.091)	(0.085)	(0.104)
N	7,660	7,646	7,660	7,221	7,429	7,660	7,219

Note: This table reports 2sCOPE estimates under alternative sets of additional controls that may partially capture latent heterogeneity. Column (a) reports the baseline specification of control variables. We then add, respectively, indicators for child wantedness, past pregnancy complications, maternal health decision autonomy, mother's age at marriage/cohabitation, and the child's age and sex. The last column includes all additional controls jointly. Sample size varies across columns due to missing values in the added covariates. Standard errors (in parentheses) are based on 1,000 bootstrap replications clustered at the regional level. Full results are available upon request. *** p < 0.01, ** p < 0.05, * p < 0.10.

Table D5. Placebo Outcomes Results

	BCG vaccine	Polio0 vaccine
Panel A. Subsample of facility births (N = 5,857)		
W	1.171***	1.171***
	(0.038)	(0.038)
ANC (γ)	-0.132	0.112
	(0.241)	(0.093)
Subsample average	0.974	0.777
Panel B. Subsample of intended pregnancy and facility births (N = 4,715)		
W	1.145***	1.145***
	(0.043)	(0.043)
ANC(γ)	-0.096	0.128
	(0.274)	(0.115)
Subsample average	0.970	0.771
All controls	yes	yes

Note: This table reports placebo tests using birth-dose vaccines (BCG and Polio 0) as outcomes. Panel A restricts the sample to facility births. Panel B further restricts the sample to intended pregnancies among facility births. The first-stage coefficient reports the effect of WHO first-trimester initiation (W) on ANC adherence from a Probit model. The second-stage coefficient reports the 2sCOPE structural effect of ANC adherence (γ). All specifications include the full set of control variables. Standard errors are in parentheses. For the 2sCOPE estimates, standard errors are based on 1,000 bootstrap replications clustered at the regional level. Full results are available upon request. *** p < 0.01, ** p < 0.05, * p < 0.10.

Table D6. Heterogeneity by maternal education level

	DTP3 vaccine		Measles vaccine	
	Frank	Clayton	Frank	Clayton
Panel A. No education				
ANC	0.443***	0.449***	0.277***	0.279***
	(0.120)	(0.123)	(0.107)	(0.103)
Panel B. Primary education				
ANC	0.787***	0.823***	0.637***	0.653***
	(0.287)	(0.301)	(0.232)	(0.219)
Panel C. Secondary and High education				
ANC	0.824	0.829	-0.464	-0.468
	(2.792)	(3.441)	(0.350)	(0.335)

Note: This table reports the structural coefficient $\boldsymbol{\gamma}$ from alternative copula specifications (Frank and Clayton). All specifications include the full set of control variables. Standard errors (in parentheses) are based on 1,000 bootstrap replications clustered at the regional level. Full results are available upon request. *** p < 0.01, ** p < 0.05, * p < 0.10.

Table D7. Results by Copula Specification for Pooled Primary and Higher Education Mothers

	Gausian	Frank	Clayton
Panel A. DTP3 vaccine			
ANC	0.779***	0.817***	0.850***
	(0.219)	(0.228)	(0.232)
Panel B. Measles vaccine			
ANC	0.381**	0.353**	0.379**
	(0.164)	(0.168)	(0.178)
Panel C. Polio 3 vaccine			
ANC	0.173	0.161	0.193
	(0.146)	(0.146)	(0.162)
Panel D. Full vaccination			
ANC	0.094	0.131	0.156
	(0.140)	(0.142)	(0.147)
Panel E. BCG vaccine			
ANC	0.079	0.090	0.150
	(0.351)	(0.336)	(0.353)
N	2,723	2,723	2,723

Note: This table reports 2sCOPE coefficient estimates for the sample pooling mothers with primary and secondary-or-higher education. Columns report results under alternative copula specifications (Gaussian, Frank, Clayton). All specifications include the full set of control variables. Standard errors (in parentheses) are based on 1,000 bootstrap replications clustered at the regional level. Full results are available upon request. *** p < 0.01, ** p < 0.05, * p < 0.10.

Table D8. Absence of Effects for Urban Girls on DTP3 vaccination

	Urban and boys		Urban and Girls	
	Frank	Clayton	Frank	Clayton
ANC	0.877**	0.954***	0.338	0.372
	(0.351)	(0.357)	(0.348)	(0.354)
N	1,138	1,138	1,172	1,172

Note: This table reports the structural coefficient γ rather than average treatment effects. All specifications include the full set of control variables. Standard errors (in parentheses) are based on 1,000 bootstrap replications clustered at the regional level. Full results are available upon request. *** p < 0.01, ** p < 0.05, * p < 0.10.

Table D9. Absence of Effects for Unintended Births

	DTP3 Vaccine: Unintended births		Measles Vaccine: Unintended births	
	Frank	Clayton	Frank	Clayton
ANC	0.306	0.390	0.025	0.039
	0.236)	(0.250)	(0.195)	(0.199)
N	1,445	1,445	1,445	1,445

Note: This table reports the structural coefficient $\boldsymbol{\gamma}$ from alternative copula specifications (Frank and Clayton) for unintended births. All specifications include the full set of control variables. Standard errors (in parentheses) are based on 1,000 bootstrap replications clustered at the regional level. Full results are available upon request. *** p < 0.01, ** p < 0.05, * p < 0.10.

Table D10. Heterogeneity Analysis by Ethnicity

	Poular Ethnicity	Others Ethnicity
Panel A. DTP3 Vaccine		
ANC	0.046	0.094***
	(0.031)	(0.019)
Panel B. Measles Vaccine		
ANC	0.030	0.080***
	(0.033)	(0.023)
All controls	yes	yes
N	2,479	5,181

Note: This table reports average treatment effects from the 2sCOPE models. All specifications include the full set of control variables. Standard errors are in parentheses and are based on 1,000 bootstrap replications clustered at the regional level. Full results are available upon request. *** p < 0.01, ** p < 0.05, * p < 0.10.

Online Appendix E. Sensitivity of Standard Errors to Clustering Level

This appendix reports 2sCOPE standard errors clustered at the primary sampling unit[20] (PSU) level rather than at the regional level.

[20] Our data include 400 PSUs.

Table E1. Robustness of Main Results to PSU-Clustered Bootstrap Standard Errors

	BCG	Polio3	DPT3	Measles	CV
Panel A. Gaussian Copula					
ANC	0.212	0.080	0.527***	0.267***	0.056
	(0.131)	(0.081)	(0.104)	(0.085)	(0.075)
Panel B. Frank Copula					
ANC	0.196	0.076	0.514***	0.273***	0.076
	(0.141)	(0.082)	(0.106)	(0.086)	(0.074)
Panel C. Clayton Copula					
ANC	0.269*	0.105	0.527***	0.287***	0.097
	(0.139)	(0.086)	(0.102)	(0.087)	(0.074)

Note: This table reports 2sCOPE estimates of the effect of ANC adherence on vaccination outcomes under alternative copula specifications (Gaussian, Frank, Clayton), with standard errors (in parentheses) based on 1,000 bootstrap replications clustered at the PSU level. Our data include 400 PSUs. Full results are available upon request. *** p < 0.01, ** p < 0.05, * p < 0.10.

Table E2. Heterogeneity by Education Level

	DTP3 vaccine			Measles vaccine		
	Gaussian	Frank	Clayton	Gaussian	Frank	Clayton
Panel A. No education						
ANC	0.449***	0.443***	0.449***	0.277***	0.277***	0.279***
	(0.120)	(0.116)	(0.112)	(0.098)	(0.099)	(0.098)
Panel B. Primary						
ANC	0.714***	0.787***	0.823***	0.598***	0.637***	0.653***
	(0.261)	(0.257)	(0.265)	(0.208)	(0.212)	(0.204)
Panel C. Secondary and High						
ANC	0.829	0.824*	0.829	-0.530*	-0.464	-0.468*
	(0.506)	(0.474)	(0.506)	(0.298)	(0.290)	(0.253)

Note: This table reports 2sCOPE estimates of the effect of ANC adherence on DTP3 and measles uptake by maternal education level under alternative copula specifications (Gaussian, Frank, Clayton). All specifications include the full set of control variables. Standard errors (in parentheses) are based on 1,000 bootstrap replications clustered at the PSU level. Full results are available upon request. *** p < 0.01, ** p < 0.05, * p < 0.10.

Table E3. Heterogeneity by Fertility Intention

	DTP3 vaccine			Measles vaccine		
	Gaussian	Frank	Clayton	Gaussian	Frank	Clayton
Panel A. Intended pregnancy						
ANC	0.562***	0.540***	0.561***	0.320***	0.326***	0.349***
	(0.118)	(0.119)	(0.117)	(0.095)	(0.098)	(0.098)
Panel B. Unintended pregnancy						
ANC	0.343*	0.306	0.390*	0.003	0.025	0.039
	(0.208)	(0.211)	(0.217)	(0.163)	(0.178)	(0.177)

Note: This table reports 2sCOPE estimates of the effect of ANC adherence on DTP3 and measles uptake by fertility intention under alternative copula specifications (Gaussian, Frank, Clayton). All specifications include the full set of control variables. Standard errors (in parentheses) are based on 1,000 bootstrap replications clustered at the PSU level. Full results are available upon request. *** p < 0.01, ** p < 0.05, * p < 0.10.

Table E4. Heterogeneity by Child Sex and Urban Residence

	DTP3 vaccine		
	Gaussian	Frank	Clayton
Panel A. Urban boys			
ANC	0.886***	0.877***	0.954***
	(0.297)	(0.298)	(0.304)
Panel B. Urban girls			
ANC	0.350	0.338	0.372
	(0.293)	(0.311)	(0.306)

Note: This table reports 2sCOPE estimates of the effect of ANC adherence on DTP3 and measles vaccination uptake by child sex in urban areas under alternative copula specifications (Gaussian, Frank, and Clayton). All specifications include the full set of control variables. Standard errors, reported in parentheses, are based on 1,000 bootstrap replications clustered at the PSU level. Full results are available upon request. *** p < 0.01, ** p < 0.05, * p < 0.10.

Online Appendix F. Test of Differential Effects

We find suggestive evidence of statistically insignificant effects among urban girls, unintended pregnancies, and Poular/Peul households. However, the difference between a significant and a non-significant estimate need not itself be statistically significant (Gelman and Stern 2006). We therefore formally test whether treatment effects differ across subgroups by estimating the difference in subgroup-specific ATEs and conducting a bootstrap test of H_0: $\widehat{ATE}_g = \widehat{ATE}_{g'}$ (e.g., urban boys vs. urban girls; intended vs. unintended pregnancies). The p-value is obtained from the empirical distribution of $\widehat{ATE}_g - \widehat{ATE}_{g'}$ using 1,000 cluster-bootstrap replications at the regional level.

Table F1. Test of Urban Gender Differences in ANC Adherence Returns (DTP3)

	Urban Boys	Urban Girls	Difference (Boys − Girls)
ATE	0.134	0.053	0.079
Bootstrap SE	(0.050)	(0.044)	(0.066)
p-value (Difference)			0.218

Note: This table reports subgroup-specific ATEs of ANC adherence on DTP3 vaccination in urban areas. The last column reports the bootstrap mean difference in ATE between urban boys and urban girls, along with its bootstrap standard error (in parentheses) and p-value from the test of equal effects.

Table F2. Test of Differences in ANC Adherence Returns by Fertility Intention

	Intended Pregnancy	Unintended Pregnancy	Difference (Intended – Unintended)
Panel A. DTP3 vaccine			
ATE	0.088	0.050	0.035
Bootstrap SE	(0.020)	(0.035)	(0.042)
p-value (Difference)			0.380
Panel B. Measles vaccine			
ATE	0.079	-0.001	0.079
Bootstrap SE	(0.023)	(0.041)	(0.045)
p-value (Difference)			0.078

Note: This table reports subgroup-specific ATEs of ANC adherence on DTP3 and measles uptake by fertility intention. The last column reports the bootstrap mean difference in ATE (Intended − Unintended), along with its bootstrap standard error (in parentheses) and p-value from the test of equal effects.

Table F3. Test of Differences in ANC Adherence Returns by Ethnicity

	Poular Ethnicity	Others Ethnicity	Difference (Others – Poular)
Panel A. DTP3 vaccine			
ATE	0.046	0.094	0.050
Bootstrap SE	(0.031)	(0.019)	(0.036)
p-value (Difference)			0.154
Panel B. Measles vaccine			
ATE	0.030	0.080	0.050
Bootstrap SE	(0.033)	(0.023)	(0.039)
p-value (Difference)			0.206

Note: This table reports subgroup-specific ATEs of ANC adherence on DTP3 and measles uptake by ethnicity (Poular/Peul vs. other ethnicities). The last column reports the bootstrap mean difference in ATE (Others − Poular/Peul), along with its bootstrap standard error (in parentheses) and p-value from the test of equal effects.

Online Appendix G. Propensity Score Matching Results

This appendix reports propensity score matching (PSM) estimates of the association between ANC utilization (four or more visits versus fewer than four) and child vaccination outcomes, adjusting for selection on observables. We implement full matching using a probit propensity score (GLM) with a caliper of 0.1. We report balance diagnostics before and after matching (Table G1) and estimated effects for each vaccination outcome under a core covariate set (column 1) and under the full covariate set used in the 2sCOPE specification (column 2) (Table G2).

Tableau G1. Balance diagnostics before and after full matching

	(1)		(2)	
	Before matching	After full matching	Before matching	After full matching
Treated units	4,006	3,996	4,006	4,006
Control units	3,654	3,654	3,654	3,654
Mean |SMD|	0.120	0.007	0.080	0.005
Max |SMD|	0.500	0.021	1.121	0.033
Share of covariates with |SMD| < 0.10	71.4	100.0	86.3	100.0
Mean propensity score, treated	0.551	0.523	0.638	0.524
Mean propensity score, controls	0.492	0.523	0.399	0.524
Common support	(0.262, 0.876)	(0.262, 0.876)	(0.066, 0.903)	(0.066, 0.903)

Note: This table reports balance diagnostics before and after full matching. Column (1) uses a core set of plausible pre-treatment covariates: mother's age, education, place of residence, number of children in the household, birth order, wealth index, and ethnicity. Column (2) uses the full set of observed covariates as in the 2sCOPE specification. SMD denotes standardized mean differences.

Table G2. Propensity Score Matching Estimates by Vaccination Outcome

	(1)	(2)
Panel A. BCG vaccination		
ANC	0.014**	0.011*
	(0.006)	(0.007)
Control mean	0.943	0.947
Treated mean	0.957	0.958
Panel B. Polio 3 vaccination		
ANC	0.012	0.013
	(0.010)	(0.012)
Control mean	0.814	0.815
Treated mean	0.826	0.828
Panel C. DTP3 vaccination		
ANC	0.050***	0.043***
	(0.008)	(0.009)
Control mean	0.877	0.881
Treated mean	0.927	0.924
Panel D. Measles vaccination		
ANC	0.053***	0.044***
	(0.010)	(0.011)
Control mean	0.800	0.809
Treated mean	0.853	0.853
Panel E. Full vaccination		
ANC	0.023**	0.024*
	(0.012)	(0.014)
Control mean	0.709	0.712
Treated mean	0.732	0.736

Note: This table reports propensity score matching (PSM) estimates of the effect of ANC adherence (four or more visits versus fewer than four) on vaccination outcomes. Column (1) uses a core set of plausible pre-treatment covariates: mother's age, education, place of residence, number of children in the household, birth order, wealth index, and ethnicity. Column (2) uses the full set of observed covariates as in the 2sCOPE specification. Robust standard errors are reported in parentheses. *** p < 0.01, ** p < 0.05, * p < 0.10.

Online Appendix H. Computation of the Average Treatment Effect in 2sCOPE

We compute the Average Treatment Effect (ATE) of antenatal care adherence on vaccination completion as:

$$\text{ATE} = \mathbb{E}\,[CV_i(1) - CV_i(0)],$$

where, $CV_i(1)$ is the potential probability of full vaccination for child i if $ANC_i = 1$ and $CV_i(0)$ is the counterfactual if $ANC_i = 0$. The expectation is taken over the empirical distribution of control variables. These potential outcomes are computed using fitted probabilities from the joint (2sCOPE) model, which estimates all structural parameters, including the copula dependence parameter $\hat{\rho}$ jointly.

Let the full vector of estimates structural parameters (excluding $\hat{\rho}$) from system (3) be denoted by $\widehat{\Theta}$. For each bootstrap replication, we generate counterfactual probabilities:

$$\widehat{P}_i(1) = CV_i(1) = \mathbb{P}\left(CV_i = 1 \mid ANC_i = 1,\ X_i,\ S_i,\ \lambda_r,\ \tau_t,\ \hat{\xi}_i^{(1)},\ W_i,\ \widehat{\Theta},\ \hat{\rho} \right),$$

$$\widehat{P}_i(0) = CV_i(0) = \mathbb{P}\left(CV_i = 1 \mid ANC_i = 0,\ X_i,\ S_i,\ \lambda_r,\ \tau_t,\ \hat{\xi}_i^{(0)},\ W_i,\ \widehat{\Theta},\ \hat{\rho} \right),$$

Here, $\hat{\xi}_i^{(s)}$ are the generalized residuals simulated under ANC_i status s $\in \{0, 1\}$. The individual-level treatment effect is computed as $\widehat{P}_i(1) - \widehat{P}_i(0)$, and the sample-level ATE is given by:

$$\widehat{ATE} = \frac{1}{n} \sum_{i=1}^{n} [\widehat{P}_i(1) - \widehat{P}_i(0)]$$

where *n* denotes the number of observations in the full sample or subsample of interest.

Robust standard errors for $\widehat{ATE}$ are obtained via 1,000 bootstrap replications clustered at the country level. In each replication, the full model is re-estimated and the ATE recalculated. The empirical standard deviation of the bootstrap distribution is reported as the standard error.